MIL-R-52805A(ME)
28 March 1980
SUPERSEDING
MIL-R-52805(ME)
9 May 1974

MILITARY SPECIFICATION

RAILWAY CARS, FLAT,

56-1/2-INCH GAGE, 300 TON,

24 WHEEL, DOMESTIC SERVICE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a heavy duty flatcar for domestic service.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-3320	- Painting and Marking: Freight and Maintenance Cars.
MIL-P-3558	- Plates, Identification: Locomotives, Railway Cars, and Work Equipment.

STANDARD

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.

FSC 2220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Code of Rules Governing the Loading of Commodities on Open Top Cars.
Code of Rules Governing the Condition of, and Repair to, Freight and Passenger Cars for Interchange of Traffic.
Specifications for Design, Fabrication and Construction of Freight Cars.
Manual of Standards and Recommended Practices.
Specification No. 2518 - Installation, Freight Car Brake Equipment.
Pamphlet No. 5039-4, Sup. 1 - Single Car Testing Device - Code of Test.
Supplement to the Manual of Standards and Recommended Practices.
Wheel and Axle Manual.
Plate B and B1
United States Safety Appliances for All Classes of Cars and Locomotives.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago, IL 60605.)

3. REQUIREMENTS

3.1 Description. The flatcar shall have a 300-ton capacity, 56-1/2-inch gage in accordance with the applicable Association of American Railraods (AAR) requirments. The underframe shall be riveted, or welded, or a combination of riveted and welded construction. Cars shall be mounted on two 12-wheel trucks with span bolsters.

3.2 First article (first-produced car). The contractor shall furnish a car for examination and test within the time frame specified to prove that his production methods will produce cars that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.8).

3.3.1 Steel. Structural steel, castings and forgings shall conform to AAR specifications.

3.3.2 Bolts, nuts and rivets. Bolts and nuts, rivets and self-locking nuts shall conform to AAR specifications.

3.4 Substitution of materials. Materials not conforming to the specifications referenced herein shall not be substituted; (1) unless such substitutions are approved; (2) when it is clearly demonstrated that an improvement

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

in operating characteristics, a savings in weight, conservation of critical or strategic materials, or a reduction in cost without sacrifice or reliability can be accomplished thereby; and (3) that such substitutions will not preclude the subsequent use of specified materials in effecting repairs or replacements. Requests for approval of such substitutions shall be submitted to contracting officer and shall be accompanined by completely detailed supporting data which clearly illustrates the use of the proposed substitute material.

3.5 Interchangeability of components. Jigs, templates, gages, and fixtures shall be used to insure interchangeability of like components and sub-assemblies with respect to assembly, performance, and quality.

3.6 Adjustments. All parts and components subject to adjustment or replacment by operating or maintnenance personnel shall be accessible. Adjusting facilities where provided shall operate without interference and shall maintain such adjustments under all conditions of service.

3.7 Tolerances and fits. Tolerances and fits shall conform to all appropriate AAR manuals, applicable to railway cars unless otherwise specified herein.

3.8 Dimensions. Nominal dimensions of cars shall be as follows:

Length over deck underframe "A"	40'-0" (Nominal) (see 6.2)
Length over deck underframe "B"	50'-0" (Nominal) (see 6.2)
Truck centers	33'-0" (Nominal)
Height of deck from top of rail at Truck centers with 216,000 lb load	4'-2" (Maximum)
Width over deck	10'-7" (Minimum) 10'-8" (Maximum)

3.9 Clearances. The cars shall not exceed the clearance limitations shown on AAR Plate B and B1.

3.9.1 Curvature. The car shall be capable of negotiating a 125-foot radius curve when uncoupled. Car shall be capable of negotiating a 125-foot radius curve when coupled by an auxiliary type coupler similar to National 31629 to a locomotive equipped with pilot type coupler (see 4.5.2.7).

3.10 Design and construction. The car shall conform to the Association of American Railroads (AAR) Manual of Standard and Recommended Practices, Codes Governing the Conditions of and Repairs to Freight and Passenger Cars for the Interchange of Traffic, and AAR Specification for Design, Fabrication and Construction of Freight Cars. Longitudinal and transverse members shall be of sufficient strength to support and transmit loads as specified in AAR Rules Governing the Loading of Commodities on Open Top Cars. Structural members and components shall have load carrying capacity to resist the design loads and superimposed dynamic loads as specified by the AAR. Structural members shall be free of notch sensitive areas and discontinuities that could result in high localized stress concentrations. The car shall be designed for the following load carrying capacity:

(a) 75 percent of the load limit concentrated on a knife edge at center of car.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

(b) 100 percent of the load limit evenly distributed between body bolsters.
(c) The car shall carry a load of 300 tons equally distributed over four load areas shown in figure 1 and 2.

3.10.1 Stability. The car shall be stabilized in accordance with D-65 of the AAR Manual of Standards and Recommended Practices, when loaded as shown in figure 2. The test load shall be Government-loaned (see 6.5).

3.11 Underframe. The all steel underframe shall be all welded, or welded-and-riveted or all riveted, covered with steel deck plating which acts as the top cover plate for center sill, side sills, and crossmembers. The four load pads as shown in figures 1 and 3 shall be raised a minimum of 50 inches above the deck surface and shall be coplanar within 25 inches at normal ambient temperatures.

3.11.1 Center sill. Center sill shall have adequate strength to support and transmit loads to the transverse members.

3.11.2 Side sills. Side sills shall be from end sill to end sill.

3.11.3 Floor stringers. Floor stringers shall be symmetrically spaced, between the center sill and side sills and shall extend between bolsters and from bolsters to end sills.

3.11.4 Crossbearers. Crossbearers shall be suitably located and of adequate strength to transfer loads to load-carrying members.

3.11.5 Crossties. Crossties shall be placed between crossbearers and bolsters and between end sills, when applicable.

3.11.6 Body bolsters. Body bolsters shall have a minimum of two web plates extending from side to center sill with deck and bottom cover plates. The car shall be equipped with jacking pads and roping staples.

3.11.7 Body center plate. Body center plate shall be AAR approved.

3.11.8 End sills. End sills shall extend from side sill to side sill.

3.12 End platform and draft sill. End plateform shall be of structural steel fabrication and suitable for mounting airbrake equipment, handbrake, and safety appliances. Decking shall be antiskid steel floot plate. Draft sill shall be fabricated from section of HSLA steel and welded to the span bolster casting. The striker and rear draft stop shall be steel castings, fabricated components or forgings welded to draft sill, and suitable for a 15-inch travel freight master end-of-car cushioning unit. All attachments shall be below deck level or shall be capable of being lowered to permit end-of-car loading.

3.13 Couplers. Couplers shall be AAR approved and shall be compatible with end-of-car cushioning couplers, shall be rotary bottom operated with coupler operating levers at diagonal corners.

3.13.1 Cusioning device. The car shall be equipped with end-of-car cushioning devices. Cushioning device shall have a 15-inch travel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

3.14 Brake system.

3.14.1 Airbrakes. Air brake shall have ABDW control valves and accessories. Airbrake installation shall be in accordance with AAR Manual Standards and Recommended Practices, Section E and Specification No. 2518 adopted by AAR, as applicable. Brake shoes shall be in accordance with the latest AAR requirements at time of contract award. The car brake piping system shall withstand an air pressure of 90 psi with a maximum pressure drop of 2 psi per minute. Brake valves and breathers shall be protected from damaging effect of road splash. Cars shall pass single car test code of test as specified in AAR leaflet 5039-4, supplement 1, or latest revision.

3.14.2 Brake force. Net airbrake and handbrake force, as determined by a static brake test shall conform to AAR requirements.

3.14.3 Airbrake hose. Airbrake hose shall conform to AAR M-601.

3.14.4 Handbrakes. Each car shall be equipped with one removable handbrake minimum of a type approved by FRA and AAR. Handbrake shall not foul the underframe or other components of the car.

3.14.5 Badge plate. A metal badge plate, cast, etched, or stamped, showing brake rigging and dimensions as specified in AAR Manual of Standards and Recommended Practices, Section E, shall be attached in an accessible location near each applicable system.

3.15 Trucks. Trucks shall be designed for a 300 ton minimum capacity car and shall be an AAR approved design. The trucks shall be equipped with hydraulic snubbing devices arranged to dampen vertical and lateral accelerations.

3.15.1 Wheels. Wheel shall be multiwear type in accordance with AAR Manual of Standards and Recommend Practices. Wheels shall be mounted in accordance with AAR requirements.

3.15.2 Axles. Axles shall be in accordance with AAR requirements. Ends of axles shall be drilled and tapped for roller bearing retaining caps.

3.15.3 Roller bearings. Roller bearings shall be AAR approved, applicable to interchange freight cars. Grease fittings shall be AAR approved pressure type.

3.15.3.1 Locking plate. Each roller bearing assembly shall include a locking plate in accordance with AAR Manual of Standards and Recommended Practices.

3.15.4 Springs. Springs shall conform to AAR M-114. The spring travel shall be compatible with hydraulic snubbing device used. The solid spring group capacity shall conform to AAR requirements.

3.16 Side bearings. Each car shall be equipped with AAR approved wear plate side bearings adjusted to AAR specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

3.17 Identification plates. An identification plate conforming to MIL-P-3558, type II, shall be attached to the car underframe in a visible location.

3.18 Painting and marking. Painting and marking shall conform to MIL-P-3320, color 14064. Reporting marks shall be DODX in lieu of USAX as specified in MIL-P-3320. Additional markings shall be in accordance with AAR specifications. Car numbers conforming to MIL-P-3320 shall be applied and shall be provided by the contracting officer (see 6.4).

3.19 Lubrication. Roller bearings shall be grease lubricated in conformance to AAR requirements and applied in quantities approved by the bearing manufacturer. Truck centerplates shall be lubricated in accordance wth AAR recommended practices.

3.20 Safety appliances. Safety appliances shall conform to United States Safety Appliances for All Classes of Cars and Locomotive in accordance with AAR, DOT or FRA where applicable. All handholds and ladder treads shall be hot-formed and rivet or boltholes shall be hot-punched or drilled. Sill steps shall conform to AAR standards.

3.21 Routing cardboard. An AAR approved routing cardboard shall be applied to each side of the car.

3.22 Placard holders. A metal placard board holder shall be applied to each side of the car and to each end of the car.

3.23 Defect cardholder. AAR type defect cardholer shall be applied to each car.

3.24 Workmanship. Workmanship for fabrication and construction shall be in accordance with AAR Specifications for Design, Fabrication, and Construction of Freight Cars. Welds shall be free of linear indications when subjected to magnetic particle inspection.

3.24.1 Welding inspection. All structural welds shall be magnetic particle inspected during welding operations. Magnetic partical inspection shall be performed on the root pass, after completion of each 1/2 inch layer and on the faces of the completed welds. In addition, the final welds shall be subjected to magnetic particle inspection after load tests, where accessible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards or for providing evidence that components have been previously approved by AAR.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) First-produced car inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced car inspection. The first-produced car which shall have been produced from production tooling shall be examined and tested to determine conformance to the requirements of this specification.

4.3.1 Examination. The first-produced car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced car shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each car shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each car shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. All cars shall be examined for compliance with the specification. This examination shall include the following:

101. Material not as specified herein.
102. Design and construction not as specified herein.
103. Dimensions, tolerances, and fits not as specified herein.
104. Parts and components missing, misalined, or not as specified.
105. Brake system and components not as specified.
106. Pipe fittings misalined, missing or not tight.
107. Adjustments not as specified.
108. Threaded parts not as specified.
109. Safety appliances not as specified herein.
110. Painting and marking not as specified.
111. Lubrication not as specified.
112. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to being tested, each car shall be lubricated with oil and grease as specified herein and put in a completely

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

operational condition. Methods of performing tests shall be submitted to the contracting officer for approval.

4.5.2.2 Airbrake. The car shall be tested with a single car testing device in accordance with Specificaion No. 2518 as adopted by AAR. Nonconformance to 3.14 shall constitute failure of this test.

4.5.2.3 Handbrake. The handbrake shall be tested by applying and releasing the brakes to determine that brake shoes are applied to all wheels and that the operation conforms to 3.14.4. Nonconformance to 3.14.4 shall constitute failure of this test.

4.5.2.4 Side bearing clearance. With the car on level tangent track and the car body level, side bearing clearance shall be measured at each bearing. Nonconformance to AAR standard limits shall constitute failure of this test.

4.5.2.5 Welds. All structural welds and at least 1/2 inch of adjacent base metal shall be magnetic particle inspected. Any evidence of linear indications shall constitute failure of this test (see 6.6).

4.5.2.6 Load. The car shall be loaded in accordance with design loads specified in 3.1 and 3.10. There shall be no permanent resulting negative camber in frame or unbalance of deck. Nonconformance to 3.1 or 3.10 shall constitute failure of this test.

4.5.2.7 Curvature. The car, uncoupled, shall be moved through a 125-foot-radius curve. At maximum swing, the brake rigging and the car body rigging shall be inspected to determine that clearances have been provided and the brake systems function as specified. A car coupled to a locomotive equipped with a pilot type coupler by means of an auxiliary coupler similar to National 31629 shall be moved through a 125-foot-radius curve to determine that the coupled units can negotiate the curve. Nonconformance to this test or 3.9.1 shall constitute failure.

4.5.2.8 Stability test. The car shall be stable when similarly tested in accordance with AAR Specification D-65, Manual of Standards and Recommended Practices. Loading characteristics shall be in accordance with figures 1 and 2. Nonconformance to AAR requirments and 3.10 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Examination. Each car shall be examined for the following defect:

113. Car not prepared for delivery as specified.

5. PACKAGING

5.1 Packaging. The cars shall be prepared for delivery in accordance with AAR requirements for interchange service.

5.2 Marking. Marking for shipment shall be in accordance with MIL-STD-129. Additional marking shall be as required by applicable AAR regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The domestic service flatcar is intended to transport special shipping containers to be secured to the car.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first-produced cars to be furnished and time frame required for submission of first-produced car(s) (see 3.2).
(c) Government-loaned property is required (see 3.10.1 and 6.5).
(d) Registration numbers to be furnished (see 3.18).
(e) Railcar underframe length (see 3.8).

6.3 First-produced car. Any changes or deviations of production cars from the approved first-produced car during production will be subject to the approval of the contracting officer. Approval of the first-produced car will not relieve the contractor of his obligation to furnish cars conforming to this specification.

6.4 Car number. The contracting officer will furnish car numbers to the contractor to be applied to each car.

6.5 Government-loaned property. The contracting officer will arrange to loan the load specified in 3.10.1.

6.6 Linear indication. Indication greater than 1/16 inch long, revealed by nondestructive test inspections, whose length is equal to or greater than three times its width. (Extracted from NAVSHIPS 0900-003-8000).

6.7 Certificate of compliance. When a certificate of compliance is submitted in lieu of a test, engineering data and computations as required for an engineering analysis shall be furnished. The certificate shall be signed by a duly authorized agent to the contractor.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 2220-A187

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

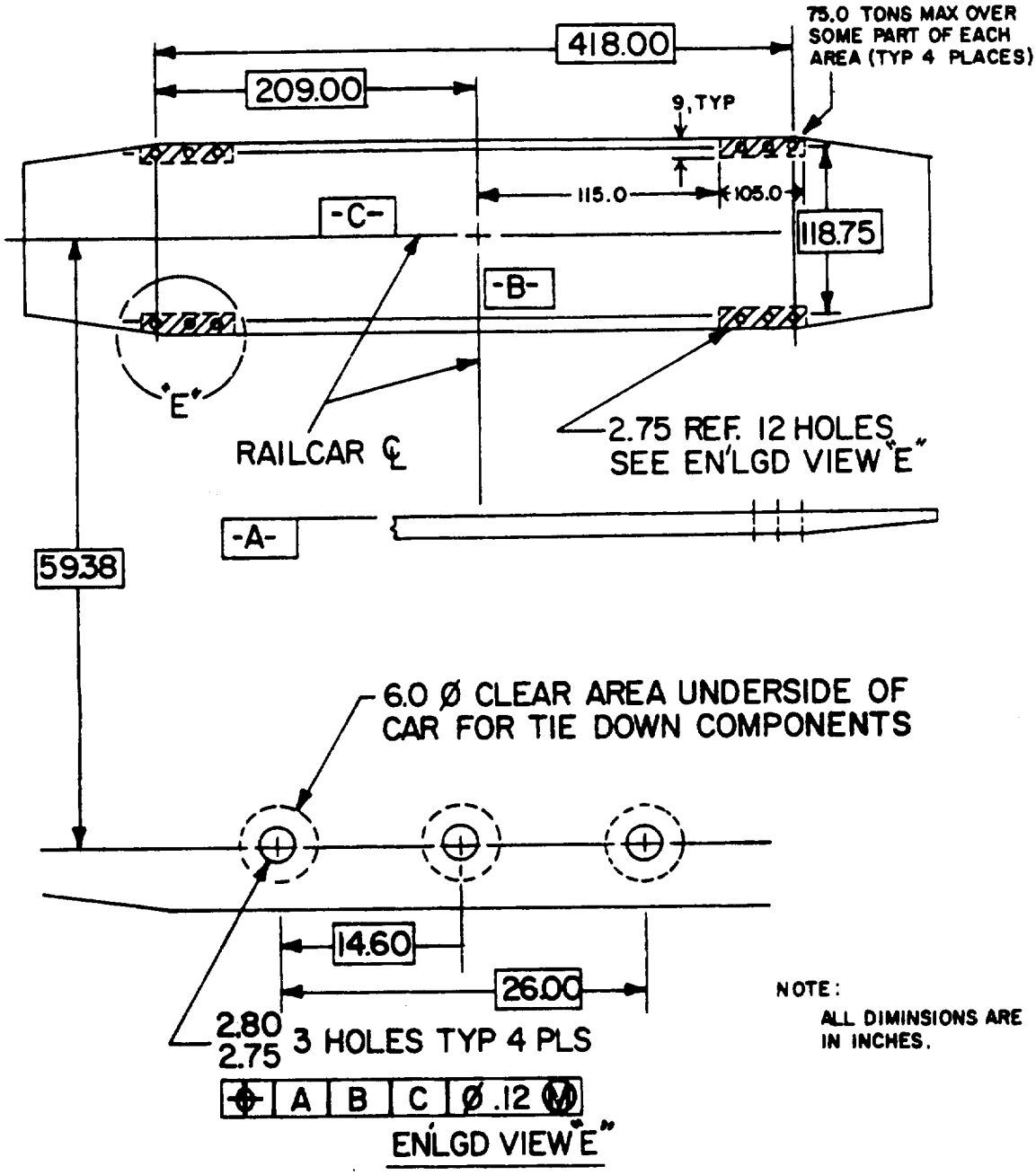
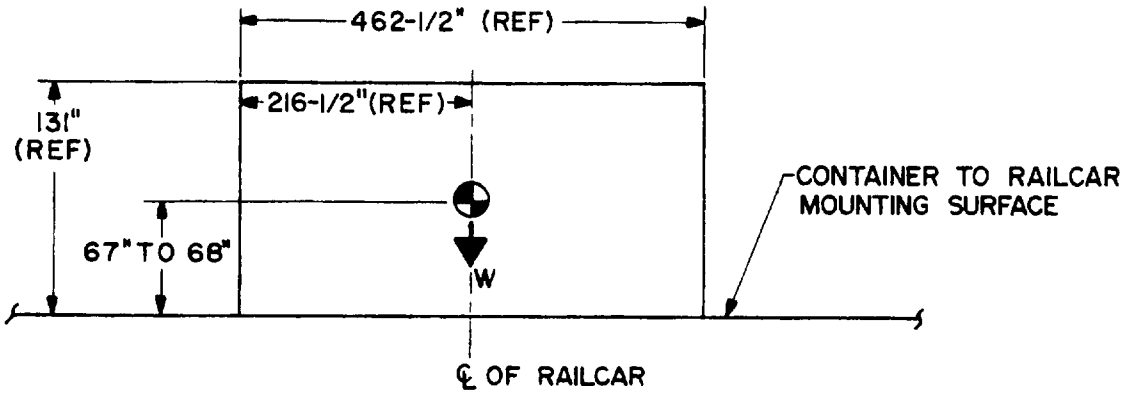


FIGURE 1.
Railcar tie down locations and cargo design loading areas.

X-3651

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.



W= 486,000 TO 540,000 LBS
WIDTH OF LOAD IS 128 INCHES (REF)

FIGURE 2. CARGO LOAD & ENVELOPE

X-2812A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-R-52805A(ME) Railway Cars, Flat, 56-1/2-Inch Gage, 300 Ton, 24 Wheel, Domestic Service

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
USA Mobility Equipment R&D Command
ATTN: DRDME-DE/DS
Ft. Belvoir, VA 22060

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.